

J Madhav Pai

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PROFESSIONAL SUMMARY

Third-year Computer Science and Business Systems undergraduate at B.M.S. College of Engineering with a strong interest in scalable software systems, backend engineering, and AI. Experienced in building end-to-end systems spanning backend architecture, data pipelines, distributed workflows, and real-world product applications.

EDUCATION

B.M.S College of Engineering, Bangalore, India

Bachelor of Engineering in Computer Science and Business Systems

CGPA: 9.4/10

Sep. 2023 – Present

TECHNICAL SKILLS

Languages: Python, C/C++, Java

AI Engineering: Hugging Face, Dataset Pipelines, Benchmark Adapters, Environment & Task Design

Agent Engineering: Agent Evaluation, MCP, Tool Calling, LangChain

Frameworks: FastAPI

Databases: PostgreSQL, MySQL

Tools: Git, Docker, Postman

CS Fundamentals: Operating Systems, OOPs, DBMS, Computer Networks

EXPERIENCE

AI Engineer Intern

Cynepia Technologies

Jan. 2026 – June 2026

Bangalore

- Designed **Trainer-bench**, a modular framework that converts heterogeneous AI benchmarks into trainable environments for agentic models.
- Standardized **10+ benchmarks** (including ADE-Bench, Deep-Planning, Spider2, and BFCL) into a unified trainable-environment format, **reducing new-benchmark integration time for developers from 2–3 days to under 45 minutes**
- Abstracted task instructions, environment state, tools, databases, and metadata to support **reinforcement learning-based agent training**.
- Implemented task validation, reproducible dataset packaging, and environment configuration to enable scalable and repeatable experimentation.
- Established foundations for RL training workflows through **episode definitions, reward hooks, and curriculum-based task sampling**.

PROJECTS

Velo – Mini-Uber | *Full-Stack Ride-Hailing Platform*

Aug. 2025 – Nov. 2025

- Built a full-stack ride-hailing platform (FastAPI, React, TypeScript, PostgreSQL, OpenStreetMap/Leaflet) **exposing 33 REST API endpoints across a relational schema of 13 tables, supporting rider, driver, and admin workflows**
- Designed a **School Pool** system for recurring student transportation with verified drivers, subscription-based passes, live ride tracking, and optimized multi-student routing.
- Implemented **OTP-based pickup and drop-off verification** to validate ride events and prevent unauthorized trip initiation during student transportation.
- Integrated **OSRM routing with Leaflet** for route calculation and interactive map visualization, enabling location-aware ride planning and driver tracking.

For-You Timeline | *Recommendation Simulator*

May 2025 – June 2025

- Simulated Twitter’s open-source **Heavy Ranker** heuristic scoring pipeline (React, Vite, Social Network Analysis), computing feed rankings from likes, retweets, replies, recency, and follow-status weights, with an interactive playground for real-time weight tuning.
- Designed a graph-based Trust Score system using Social Network Analysis, propagating trust from verified “seed” institutional nodes across the network via a custom weighted-edge decay model.
- Built a “Kid-Safe Mode” content-safety feature that filters low-trust content in real time using the propagated trust graph, downranking bad actors without relying on keyword-based filtering.
- Built a custom force-directed graph visualization engine from scratch — **no external charting libraries** — to render trust propagation and network dynamics interactively in the browser.